ARC-AGI-3

A Detailed Briefing on the Interactive Reasoning Benchmark
and Research Landscape for a Final Year Project

Prepared for FYP Scoping — August 2026

This document summarizes ARC-AGI-3 (the third generation of the Abstraction and Reasoning Corpus, launched by the ARC Prize Foundation on March 25, 2026), its technical design, scoring methodology, the associated ARC Prize 2026 competition, the current state of the art, and a curated set of academic references and candidate research directions suitable for a final-year project.

1. Background: The ARC-AGI Series

The Abstraction and Reasoning Corpus (ARC) was introduced by François Chollet in his 2019 paper "*On the Measure of Intelligence*" as a benchmark designed to measure **fluid intelligence** and **skill-acquisition efficiency** rather than task-specific skill. Its central thesis is that intelligence should be measured by how efficiently a system can acquire new skills for genuinely novel tasks, not by how well it performs on tasks it may have memorized or been trained on. ARC tasks are deliberately built only from "Core Knowledge priors" — basic human cognitive abilities such as objectness, counting, symmetry, topology, and elementary physics — so that no specialized domain knowledge or language understanding is required to solve them.

Version	Release	Format	Primary Focus	Best reported AI score (approx., at time of writing)
ARC-AGI-1	2019	Static grid input–output pairs (few-shot, ~3 examples per task)	Fluid intelligence / abstraction from minimal examples	~45% (Tiny Recursive Model, 2025); frontier LLM systems have separately claimed higher scores under heavy test-time compute
ARC-AGI-2	March 2025	Static grid input–output pairs, harder / more compositional tasks, tighter human-calibrated difficulty	Resistance to brute-force test-time compute scaling; finer-grained signal at higher cognitive complexity	~24% on the private evaluation set (ARC Prize 2025, top team NVARC)
ARC-AGI-3	March 25, 2026	Interactive, turn-based grid *environments* (games), no input–output pairs, no instructions	Exploration, world-model building, goal inference, and continuous learning over time	<1% of the efficiency-normalized human baseline (frontier models, March 2026); ~0.25–12.6% on preview tasks depending on approach

ARC-AGI-1 and ARC-AGI-2 share the same static format: a handful of example input/output grid pairs are given, and the system must infer the transformation rule and apply it to a new test input. ARC-AGI-3 is a structural break from this format — it moves from "solve this puzzle" to "learn to play this game," which is why the ARC Prize team describes it as the first genuinely **agentic** and **interactive** benchmark in the series.

2. What Is ARC-AGI-3?

ARC-AGI-3 is described by its creators as **"the first interactive reasoning benchmark designed to measure human-like intelligence in AI agents."** Instead of presenting static examples, it drops an agent into a novel, handcrafted, turn-based environment (structurally similar to a simple 2D video game) with **no instructions, no stated rules, and no explicit goals**. The agent must:

- **Explore** the environment to discover how it behaves;
- **Build an internal world model** from its observations;
- **Infer the goal** on its own, from implicit feedback (e.g. score changes, level transitions) rather than being told what to do; and
- **Plan and act** over a long horizon with sparse feedback, adapting its strategy as it learns more.

> **Design philosophy.** As stated in the official launch announcement: "*There are no instructions, no rules, and no stated goals.*" The benchmark deliberately avoids language, numbers, and cultural symbols so that success cannot come from memorized world knowledge or instruction-following — only from Core Knowledge priors (objectness, agency, geometry/topology, elementary physics) applied through direct interaction.

2.1 Task / Environment Structure

- Each environment is a turn-based "game" rendered on a grid of up to **64×64 cells**, with up to **16 possible colors** per cell — the same visual grammar as the static ARC-AGI puzzles, but now dynamic and responsive to actions.
- On each turn, the agent submits **one action** (e.g. directional moves, discrete "key press" style actions, or grid-coordinate clicks), and the environment updates and returns a new grid state.
- Environments are organized into **difficulty-ordered levels** within each game — typically starting with an implicit "tutorial" level and escalating in complexity.
- At full launch the benchmark comprises **135 total environments** (hundreds of environments were originally targeted, with thousands of individual levels), split into public, semi-private, and fully private sets at roughly a **1:5 public-to-private ratio** — a deliberate change from ARC-AGI-1/2's ~1:10 ratio, and from the "only 25 public" figure reported around launch, to reduce the risk of environments leaking into training data.
- All environments are handcrafted by an in-house team of human game designers and are extensively difficulty-calibrated using real human playtesters before release.

2.2 Technical / SDK Details

ARC Prize provides an open-source Python toolkit ("ARC-AGI Toolkit") and REST API for building and evaluating agents:

```
pip install arc-agi          # or: uv add arc-agi

import arc_agi
arc = arc_agi.Arcade()
env = arc.make("ls20", render_mode="terminal")

env.step(GameAction.ACTION1)   # submit one action per turn
arc.get_scorecard()            # retrieve results / metrics
```

Omitting `render_mode` substantially increases throughput (reportedly >2000 FPS) for large-scale training runs. Full documentation, environment lists, and agent quick-start templates are published at docs.arcprize.org and arcprize.org/tasks.

2.3 Scoring Methodology — the Efficiency Metric

Unlike ARC-AGI-1/2 (binary pass/fail per task), ARC-AGI-3 scores agents on **efficiency relative to a human baseline** — the ARC Prize team frames this as "intelligence is efficiency." The reported per-level formula is approximately:

$$\text{Level Score} = \min\left(1.0, \left(\frac{\text{human_baseline_actions}}{\text{agent_actions}}\right)^2\right)$$

- The **squared** term heavily penalizes inefficient / brute-force solutions: an agent using 10× the human's action count scores near zero on that level, and a hard cutoff at 5× the baseline yields a score of exactly zero.
- The human baseline for each level is taken as the **second-best** of 10 first-time human testers (filtering out lucky outliers while keeping the bar grounded in genuine human performance).
- Later levels within a game are weighted more heavily than earlier ones (Level 1 = weight 1, Level 2 = weight 2, ...), so trivial tutorial levels barely move the overall score.
- A perfect score of **100%** means the agent matches human efficiency across every game and level.
- The system also logs full **replays** and structured decision timelines for transparency and post-hoc analysis of agent reasoning — a rich source of data for research.

3. The ARC Prize 2026 Competition

ARC-AGI-3 is the centerpiece of the **ARC Prize 2026** competition, hosted on Kaggle, with a total prize pool of roughly **US$2,000,000** across tracks (the ARC-AGI-3 track alone is **US$850,000**).

Prize category	Amount	Condition
Grand Prize	$700,000	First eligible agent to reach 100% on the evaluation set; rolls over if unclaimed
Top Score Awards	$75,000 total ($40K/$15K/$10K/$5K/$5K for places 1–5)	Highest scores at final deadline
Milestone Prizes	$75,000 total, split across two checkpoints	June 30, 2026 and September 30, 2026; requires open-sourcing the solution at that deadline

3.1 Rules and Timeline

- Submissions are made via **Kaggle**; evaluation sandboxes have **no internet access** (this blocks calling hosted/proprietary LLM APIs at inference time).
- **All prize-eligible code and methods must be open-sourced** under a permissive license (e.g. CC0 / MIT-0 equivalent) at the relevant deadline.
- Milestones: **June 30, 2026** and **September 30, 2026**; final submission deadline **November 2, 2026**; results announced **December 4, 2026**.
- Agents are evaluated on four simultaneously-tested capabilities: **Exploration** (active information gathering), **Modeling** (building a generalizable world model from observations), **Goal-setting** (autonomously inferring desirable end-states without being told), and **Planning & Execution** (strategic, adaptive action sequencing).

3.2 The 30-Day Preview: What Happened

Before the full launch, ARC Prize ran a 30-day public preview using three sample environments (`ls20` – agentic/map-based symbol navigation, `ft09` – logic-based pattern matching, and `vc33` – multi-object volume orchestration). Key results:

Rank	Agent	Score (% of human baseline)	Approach
1st	StochasticGoose	12.58%	CNN-based reinforcement learning
2nd	Blind Squirrel	6.71%	State graph with a ResNet value model
3rd	Explore Till You Solve It	3.64%	Frame-graph analysis

Notably, the top preview agent's performance did **not generalize** to the full private benchmark at launch (StochasticGoose's score reportedly dropped to ~0.25% on the complete, harder benchmark), while frontier commercial AI systems scored roughly **0.51%** in aggregate as of March 2026 — versus **100%** for human testers (1,200+ players, 3,900+ games played during the preview). This is the headline "AI vs. human" gap the benchmark is built to expose, and the ARC Prize team explicitly notes that "progress will likely come from better agent scaffolding or new architectures, not task-specific tuning."

4. State of the Art & Context from ARC Prize 2025

The ARC Prize 2025 Technical Report (covering the ARC-AGI-2 competition immediately preceding ARC-AGI-3's launch) is useful context: it attracted 1,455 teams, the top score on the ARC-AGI-2 private evaluation set was **24.03%**, and it identified **"refinement loops"** — iterative program optimization guided by feedback — as the dominant emerging technique. Representative approaches:

Approach	Key idea	Reported result
NVARC (1st place, ARC Prize 2025)	Test-time training with synthetic data generation	24.03% on ARC-AGI-2 private set
Tiny Recursive Model (TRM)	~7M-parameter recursive reasoning network, no pretraining	~45% on ARC-AGI-1
CompressARC / "ARC-AGI Without Pretraining"	~76K-parameter model, MDL-based, pure test-time optimization	~20% on ARC-AGI-1
SOAR	Self-improving evolutionary program synthesis guided by an LLM	Large relative improvement on ARC-AGI-1 (reported ~52% gain over baseline)

The report also flags a methodological concern relevant to FYP work: evidence of **benchmark contamination**, where frontier models appear to reproduce ARC's specific color-mapping conventions despite no explicit training on them, suggesting ARC-AGI puzzle data (or close derivatives) is present in some public pretraining corpora. This is one of the reasons ARC-AGI-3's fully interactive, harder-to-memorize format was introduced.

5. Curated Academic & Technical References

All links verified via web search in August 2026. Where a paper has both an arXiv abstract page and PDF, the abstract page is listed.

5.1 Foundational / Benchmark Papers

Core **Chollet, F. (2019). On the Measure of Intelligence.**
arXiv:1911.01547 — arxiv.org/abs/1911.01547. The original paper introducing ARC and the theoretical framework (fluid intelligence, skill-acquisition efficiency, Core Knowledge priors) underlying the entire ARC-AGI series.

Core **Chollet, F. et al. ARC-AGI-2: A New Challenge for Frontier AI Reasoning Systems.**
arXiv:2505.11831 — arxiv.org/abs/2505.11831. Describes the design goals, human-calibration methodology, and results motivating the move from ARC-AGI-1 to ARC-AGI-2.

Core **ARC-AGI-3: A New Challenge for Frontier Agentic Intelligence.**
arXiv:2603.24621 — arxiv.org/abs/2603.24621. The technical paper accompanying the ARC-AGI-3 launch; covers the interactive benchmark design, Core-Knowledge-only environment construction methodology, human-subject calibration, and initial frontier-model results.

Core **ARC Prize 2025: Technical Report.**
arXiv:2601.10904 — arxiv.org/html/2601.10904v1. Competition results, top approaches ("refinement loops"), contamination analysis, and the rationale for ARC-AGI-3.

ARC Prize 2024: Technical Report.
arcprize.org/media/arc-prize-2024-technical-report.pdf. Earlier competition cycle; useful for tracking year-over-year methodology shifts.

A Comprehensive Behavioral Dataset for the Abstraction and Reasoning Corpus.
Scientific Data (Nature) — nature.com/articles/s41597-025-05687-1. Large human behavioral dataset on ARC tasks; useful for building human-baseline comparisons.

5.2 Program Synthesis & Symbolic / DSL Approaches (ARC-AGI-1/2)

Program Synthesis using Inductive Logic Programming for the Abstraction and Reasoning Corpus.
arXiv:2405.06399 — arxiv.org/html/2405.06399v1

First Steps of an Approach to the ARC Challenge based on Descriptive Grid Models and the Minimum Description Length Principle.
arXiv:2112.00848 — arxiv.org/abs/2112.00848

ConceptSearch: Towards Efficient Program Search Using LLMs for ARC.
arXiv:2412.07322 — arxiv.org/html/2412.07322v2

How to Beat ARC-AGI by Combining Deep Learning and Program Synthesis (ARC Prize blog / write-up of the 2024 winning approach).
arcprize.org/blog/beat-arc-agi-deep-learning-and-program-synthesis

5.3 Neural / Test-Time-Training Approaches (2025–2026)

Less is More: Recursive Reasoning with Tiny Networks (Tiny Recursive Model).
arXiv:2510.04871 — huggingface.co/papers/2510.04871; follow-up analysis: *Tiny Recursive Models on ARC-AGI-1: Inductive Biases, Identity Conditioning, and Test-Time Compute*, arXiv:2512.11847 — arxiv.org/abs/2512.11847. Code: github.com/SamsungSAILMontreal/TinyRecursiveModels.

Liao, I. et al. ARC-AGI Without Pretraining (CompressARC).
arXiv:2512.06104 — arxiv.org/abs/2512.06104. A ~76K-parameter model solving ARC-AGI-1 tasks purely via test-time optimization / minimum description length, with no pretraining at all — a strong reference point for compute-constrained FYP work.

Pourcel, J., Colas, C., Oudeyer, P-Y. Self-Improving Language Models for Evolutionary Program Synthesis: A Case Study on ARC-AGI (SOAR).
arXiv:2507.14172 — arxiv.org/abs/2507.14172 (ICML 2025).

5.4 Agentic / Reinforcement-Learning & World-Model Approaches (most relevant to ARC-AGI-3's interactive format)

Reinforcement World Model Learning for LLM-based Agents.
arXiv:2602.05842 — arxiv.org/abs/2602.05842

Toward Efficient Exploration by Large Language-Model Agents (ICLR 2026).
dilipa.github.io/papers/iclr26_psrl_llms.pdf

Meta-RL Induces Exploration in LLM Agents (ICLR 2026).
openreview.net (id=4GiBscHW1k)

WorldCompass: Reinforcement Learning for Long-Horizon World Models.
ResearchGate

SIGHT: RL with Self-Evidence and Information-Gain Diverse Branching for Search Agents.
arXiv:2602.11551 — arxiv.org/html/2602.11551v1

5.5 Primary Sources / Living Documents (non-peer-reviewed but essential)

ARC-AGI-3 official benchmark page.
arcprize.org/arc-agi/3

Announcing ARC-AGI-3 (launch blog post).
arcprize.org/blog/arc-agi-3-launch

ARC-AGI-3 Preview: 30-Day Learnings.
arcprize.org/blog/arc-agi-3-preview-30-day-learnings

ARC Prize 2026 – ARC-AGI-3 Competition (rules, prizes, timeline).
arcprize.org/competitions/2026/arc-agi-3

ARC-AGI-3 developer documentation (SDK / API).
docs.arcprize.org

ARC-AGI-3: The New Interactive Reasoning Benchmark (explainer with comparison tables).
datacamp.com/blog/arc-agi-3

ARC-AGI Prize 2025 Results and Analysis.
arcprize.org/blog/arc-prize-2025-results-analysis

ARC-AGI-3 Leaderboard & Scores (community-tracked, updated).
benchlm.ai/benchmarks/arcagi3

6. Candidate FYP Directions

Given that ARC-AGI-3 is brand new (launched March 2026) and current AI performance is still extremely low (well under 1% of human efficiency on the private set), there is substantial open ground. Some directions worth considering, roughly ordered from most tractable for a solo FYP to most ambitious:

6.1 Empirical / analysis-oriented (lower compute, strong novelty)

- **Failure-mode analysis of existing agents.** Use the public preview environments and replay/timeline data to systematically categorize *why* agents like StochasticGoose or Blind Squirrel fail (e.g. poor exploration vs. poor goal inference vs. poor long-horizon planning) — this only requires the SDK and public games, not a novel agent.
- **Reproducing and extending the RHAE efficiency metric** on a custom set of small hand-built environments, to study how metric design shapes what "intelligence" gets rewarded.
- **Benchmark design study:** build a small number of new ARC-AGI-3-style environments and run controlled human studies (a scaled-down version of what ARC Prize itself does), analyzing which design choices make environments resistant to brute-force/random search — directly extending the "vulnerable to brute force" finding from the 30-day preview.

6.2 Agent-building (core ML/RL project)

- **Model-based RL with explicit world models** (e.g. a learned transition model + planning, in the spirit of Dreamer-style architectures) applied to the public ARC-AGI-3 environments, testing whether explicit world-model learning outperforms model-free CNN/ResNet baselines like the current leaders.
- **Curiosity-driven / intrinsic-motivation exploration** (e.g. prediction-error or information-gain bonuses) to address the "no stated goal" exploration problem specifically.
- **Vision-language-model agent scaffolding:** use a multimodal LLM as a high-level planner (perceiving grid states as images) combined with a lower-level action policy, and study how scaffolding design affects sample efficiency — directly relevant to the ARC Prize team's own hypothesis that "progress will come from better agent scaffolding."
- **Transfer from ARC-AGI-1/2 program-synthesis techniques** (DSL search, MDL-based compression like CompressARC) into the interactive setting, e.g. by having an agent synthesize a candidate "rule program" for the environment from observed transitions and using it for planning.

6.3 More ambitious / research-novel

- **Meta-learning across environments:** train an agent on a subset of public ARC-AGI-3 games and measure few-shot adaptation speed on held-out games, as a direct proxy for the benchmark's "skill-acquisition efficiency" thesis.
- **Neuro-symbolic hybrid agents** combining a perception module (objectness/segmentation over the grid) with a symbolic planner operating over inferred object/rule representations.
- **Contamination-resistant evaluation methodology:** building on the ARC Prize 2025 report's contamination findings, design and test methods for detecting whether an LLM-based agent has memorized parts of a public environment versus genuinely reasoning about it.

> **Practical note for scoping.** The ARC-AGI-3 SDK (`pip install arc-agi`) and public environments are freely accessible now, evaluation/replay tooling is already built for you, and the field is early enough that even a modest, well-evaluated agent or a rigorous empirical analysis can be a genuine contribution. Aligning your FYP with one of the June 30 or September 30, 2026 ARC Prize milestones could also give you an external, dated benchmark for your own results and, if your supervisor allows external competition participation, a route to publish/open-source your work alongside the official competition.